BLUEPRINT - WEEK 11 - TEAM 3

When Scarcity Ends - And What We Lose Anyway | Chapter 10: Dreaming of Plenitude

FACTS WE AGREE ON

• Water scarcity reflects infrastructure, distribution, information, affordability, and operational constraints - not visibility alone.

• Water sensors, leak alerts, and monitoring platforms already exist; new technology must offer incremental value.

• Detection does not establish repair, and a closed ticket does not independently prove reduced water loss.

• Household and network tools can exclude underserved communities if they require costly hardware, reliable connectivity, or institutional capacity.

BETS WE CHOSE

• **Team-confirmed vote: 3-2 for Loss Visibility** (Money, Operator, Adversary); User and Technologist voted for Household Water.

• Focus on the path from suspected network leak to inspection, assignment, physical repair, and verified outcome, rather than new sensing technology.

• Use human-reviewed repair prioritization balancing estimated loss, service interruption, feasibility, and community vulnerability.

• Adapt proven global approaches (TaKaDu, ASTERRA, FIDO, Singapore PUB, Manila Water) to low-cost local operations; no claim of proven Mexican pilot results.

PRIMARY VACUUM

• **Loss Visibility:** the operational gap between identifying possible municipal water-network leaks and completing documented, verified repairs. **Backup:** Household Water planning under uncertainty.

THE CONDITIONS

1. **Actionable closure:** every event has an owner, next action, status, and resolution record.

2. **Human authority:** no automated excavation, structural diagnosis, or major dispatch without qualified human verification.

3. **Incremental evidence:** compare with existing workflows; measure delays, costs, verified repairs, and actual loss reduction where measurable.

4. **SHADOW CLAUSE - equity and anti-rationing:** include service vulnerability in human-reviewed priorities; no pay-to-prioritize access, punitive household profiling, or discriminatory automated rationing.

5. **Local feasibility:** use existing data and familiar low-cost workflows first; justify each added sensor, AI component, and connectivity dependency.

6. **Truth and auditability:** distinguish suspected, physically confirmed, repaired, and outcome-verified events; label uncertainty and preserve human correction.

DECLARATIONS

• **Monica (USER)** - Household-facing status viewer distinguishing verified repairs from unconfirmed reports (conditions 4, 6).

• **Ana Maria (TECHNOLOGIST)** - Compare basic spreadsheet tracking with rule-based repair prioritization on labeled hypothetical cases (3, 6).

• **Money (name to confirm)** - Sensitivity model of repair economics, avoided losses, and equity tradeoffs (3, 4).

• **Jose Luis (OPERATOR)** - AquaRail minimum ticket workflow from suspected leak to human inspection, repair, and outcome record (1, 2, 5).

• **Diego Gil (ADVERSARY)** - Adversarial tests for biased prioritization, unresolved tickets, and false success labels (2, 4, 6).

DISSENT PRESERVED

• Monica and Ana Maria: municipal repair capacity makes household planning a more directly actionable alternative. Money: equity weighting could weaken financial payback without a funded mechanism. Diego: even a perfect workflow fails if repair capacity and equitable allocation are missing.

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